

Ritika Shrestha

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EDUCATION

Denison University

Granville, OH

B.S in Computer Science and Mathematics, GPA: 3.79/4.00

May 2026

- *Coursework:* AI, Time Series Analysis, Stat Modeling, Real Analysis, Operating Systems, Computer Systems, Database
- *Awards:* Forbes B. Wiley Award (Math), Dean's List

Cornell Tech

May 2025 - May 2026

Breakthrough Tech **AI/ML**, Fellow

EXPERIENCE

Handshake AI, Fellow

Handshake

April 2026 - Present

- Ship **agentic** coding projects end-to-end from natural-language specs, owning a 5-phase GitHub workflow to scope tasks, write evaluation rubrics, and benchmark 4 frontier models against them

Junior Software Engineer

Denison IT Web Services

Nov 2022 - May 2026

- Built a customizable app dashboard for the university's internal portal using **Vue.js** inside **Drupal Twig** with drag-and-drop layouts, filtering, and saved per-user preferences; Google Analytics showed a **25%** increase in engagement
- Automated build and deployment workflow in **GitLab CI/CD** and resolved production support tickets over 4 years

Software Engineering Intern

TransCore

June 2025 - Aug 2025

- Shipped a **RAG** chatbot that automated FAQ and toll-processing inquiries for customer service reps, hosting quantized **Llama 3.2 Vision**, Mistral, and Qwen locally via **Ollama** to keep customer data in-house within **GPU** memory limits
- Applied **LoRA fine-tuning** for tone-aware, de-escalating responses and built security **guardrails**, including input sanitization and semantic routing to block prompt injection before it reached the model

Software Engineering Intern

Real AI Dynamics (RAID)

Dec 2024 - May 2025

- Built a **collaborative-filtering** recommendation service for micro-SaaS platforms, serving recommendations across **10** products

Teaching Assistant

Denison Computer Science Department

Jan 2024 - May 2024

- Tutored CS students and co-designed course structures with faculty, contributing to a **10%** improvement in student grades

PERSONAL PROJECTS

Little AI Helper — Published on the Chrome Web Store

April 2026 - Present

- Built and published an AI-powered **Chrome extension** that autofills job applications, generates tailored answers and cover letters, and flags visa-sponsorship requirements, powered by **Gemini** and Cloud Run with **Vertex AI** proxy

Blueberry Mart App — In closed testing on Google Play

Jan 2026 - Present

- Building the digital infrastructure for a multi-branch grocery mart launching soon in Nepal using **.NET 8**, **React Native**, **Vite** with role-based access and automated EF Core 8 migrations on **PostgreSQL/Cloud SQL**, deployed to **Cloud Run**
- Engineered a real-time inventory analytics pipeline using Kafka and **BigQuery** alongside integrations for eSewa payments, tiered memberships, and a self-service query builder for shareholders

Jobbie Application — Microservice job posting platform

Dec 2024 - Dec 2025

- Led a team of 5 to build 5 backend services (**C#**, **.NET 8**, **Node.js**) communicating over **gRPC**, HTTP, and **Kafka**; containerized with **Docker** and deployed on **Azure** Kubernetes Service with GitHub Actions CI/CD
- Replaced synchronous **REST** calls between services with Kafka events after request chains began blocking on slow downstreams, and used the **transactional outbox** pattern to guarantee event delivery across **MySQL** and **SQL Server**

Flavor Radar (Nestlé)

Aug 2025 - Dec 2025

- Built a sentiment analysis pipeline on **14M Amazon reviews**, benchmarking three models, **RoBERTa**, Logistic Regression (**TF-IDF**), and VADER baseline, lifting **macro-F1** from **0.54** to **0.74+** across all sentiment classes
- Engineered a composite scoring system blending **popularity**, **growth velocity**, and **sentiment** to surface emerging flavor trends (Whole Bean Coffee, Cookies & Cream, Caramel), enabling data-driven R&D decisions for **Nestlé**

TECHNICAL SKILLS

- **Languages:** C++, C#, Python, SQL, TypeScript, JavaScript, PHP
- **Technologies:** .NET, Django, React, React Native, Node.js, Next.js, MongoDB, AWS, TensorFlow, Kubernetes, gRPC, RabbitMQ, Azure, Arduino, GCP

CAMPUS INVOLVEMENT

Officer, OH

Women in Math and Computer Science

Oct 2022 - May 2026

- Organized events and represented the org at the **Ohio Celebration of Women in Computing (2023)**